Examiner only

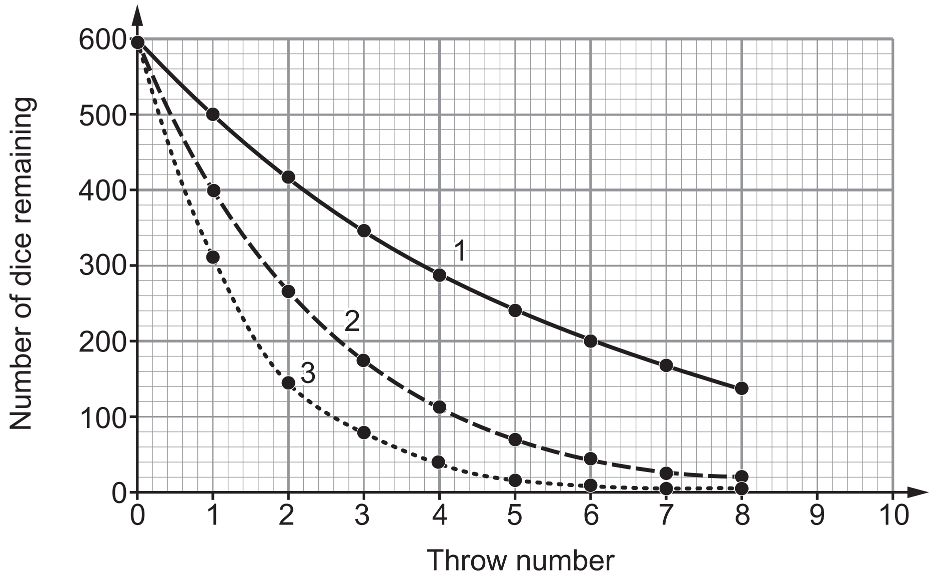
9. The following questions are based on the information on pages 6 and 7 of the Resource Folder about using dice to model half-life.

(a) Each group's results were added together to give class results. Give **one** reason why this makes the data more repeatable. [1]

....

....

(b) The results in **Figures 7 and 8** are used to plot graphs on the grid below.



(i) Add lines to the graph to determine the half-life represented by line 3. [2]

half-life = ... throws

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(ii) Use the information in the graph to tick (✓) the boxes next to the **three** correct statements. [3]

Line 1 represents the data from Model 2.	☐
Line 2 shows decay of the quickest rate.	☐
Line 2 shows the equivalent of three half-lives after 5 throws.	☐
Line 1 represents the longest half-life.	☐
Line 3 would have been produced if 600 coins had been used instead of dice and the number of 'heads' counted after each throw.	☐
Line 3 has an activity that is 1/8 of the original after 8 throws.	☐

(c) The teacher suggests to the class that they add 5 red wooden cubes to the 50 dice they start with. The red cubes would not be removed after each throw but would be counted every time to improve the modelling of radioactive decay.

(i) State what the red cubes represent. [1]

....

(ii) State how the presence of the red cubes would affect line 1 on the graph. [1]

....

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END OF PAPER